



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

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February 13th, 2025

About Quiz

PHY1001 Dummy Quiz



长按图片扫码

When is the first quiz for PHY1001 L2?

- A: Next Monday
- B: Next Tuesday
- C: Next Wednesday
- D: None of above

- Scan QR code with either WeChat or phone camera
- You will need to fill
 - Name (first, last): i.e. Jingqiang Ye
 - Student ID: xxxxxxxxxx
 - Answer for the single-choice question.
- Please only submit once.
 - Only the last attempt will be accounted for if you submit multiple times.
- The first quiz will be from lecture 1 - 6.

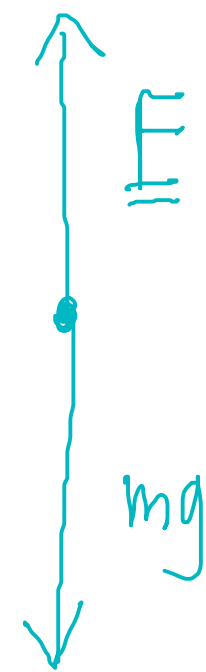
Human Power



Red Rocks, Nevada

3 sig. fig.

A person of $m=50.0$ kg climbs up a cliff of $h=4.00 \times 10^2$ m in 15.0 minutes! What is the average power output?



$$W = F \cdot h$$

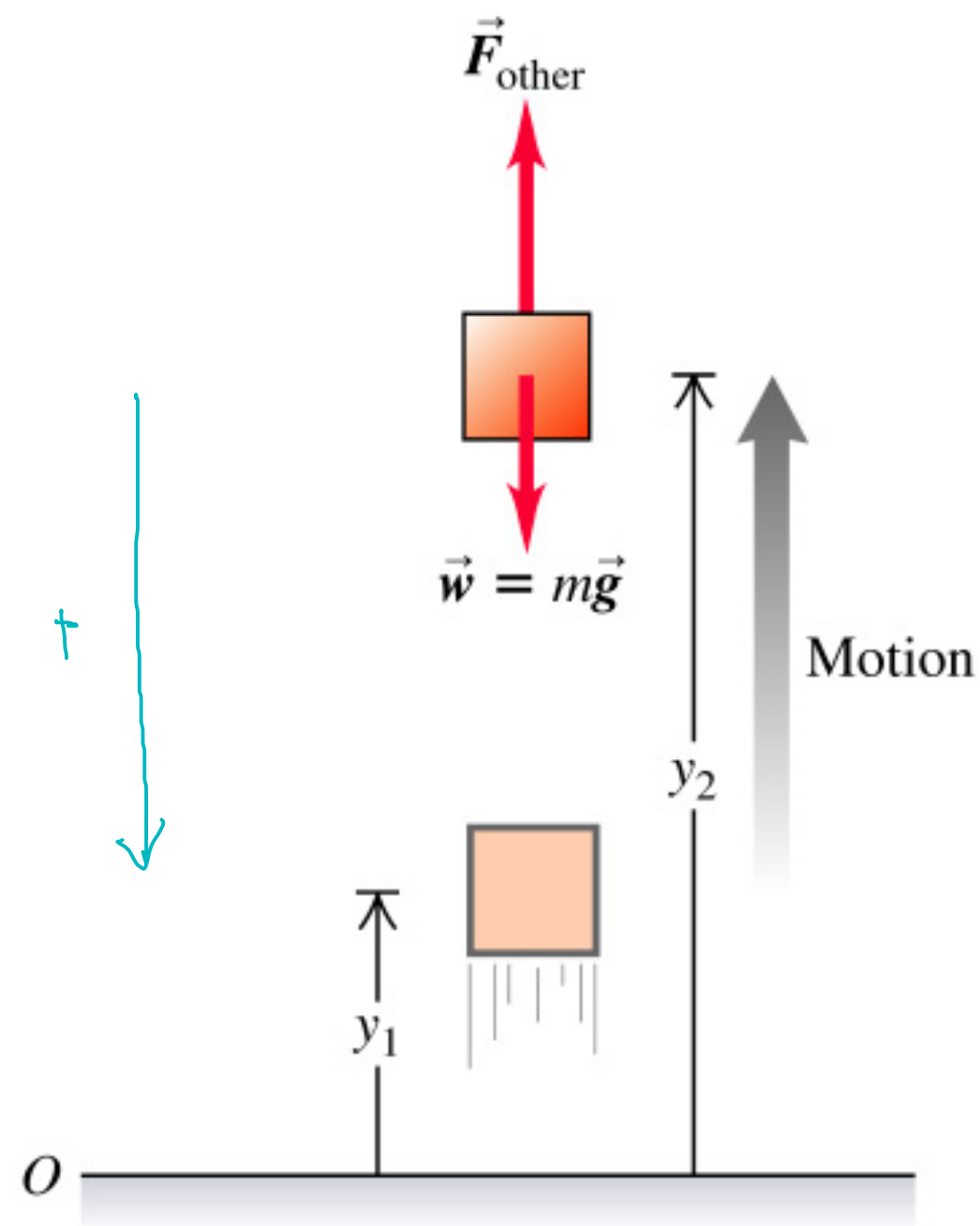
$$F = mg$$

$$W = mgh$$

$$P = \frac{mgh}{t} = \frac{50 \cdot 9.8 \cdot 400}{15 \cdot 60} \text{ W}$$

$$= 218 \text{ W}$$

Gravitational Potential Energy



- Gravitational potential energy $U_g = mgh$, h is the height above the reference point.
- Unit of (potential) energy: J.
- $\Delta U_g = -W_g$, when W_g is the work done by gravity

$$\begin{aligned}
 W_g &= mg(-y_1 - y_2) = -mgy_2 - mgy_1 \\
 &= -(U_2 - U_1) \\
 &= -\Delta U
 \end{aligned}$$

Revisit the Question

A kid of mass m sits on a swing with a radius R . An adult pushes the kid by exerting horizontal force F that starts at 0 and gradually increases such that the swing moves very slowly and always remains in equilibrium. In the end, the angle of the swing is θ_0 w.r.t vertical.

c) Find the work done by the adult.

a)
b)

All forces on the kid

the tension T of the swing on the kid

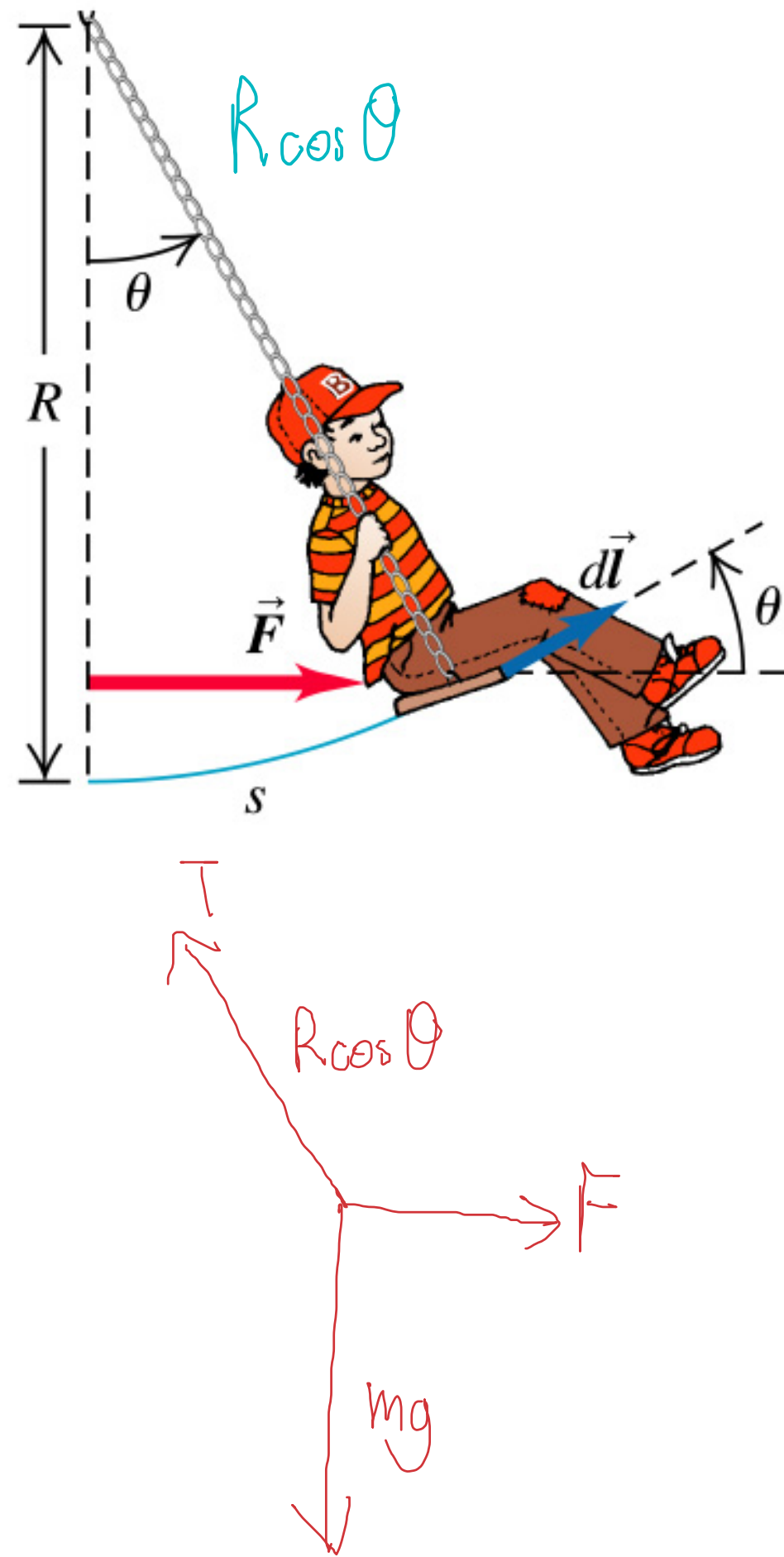
a) $W-k$ $W_{\text{all}} = \Delta k = 0$

b) $W_i = 0 = \int \vec{T} \cdot d\vec{l}$

c) Last lesson: $W_F = \int \vec{F} \cdot d\vec{l} = \int mg \tan \theta \cdot R d\theta \cdot \cos \theta$

c) $W_{\text{all}} = W_F + W_{mg} = 0$

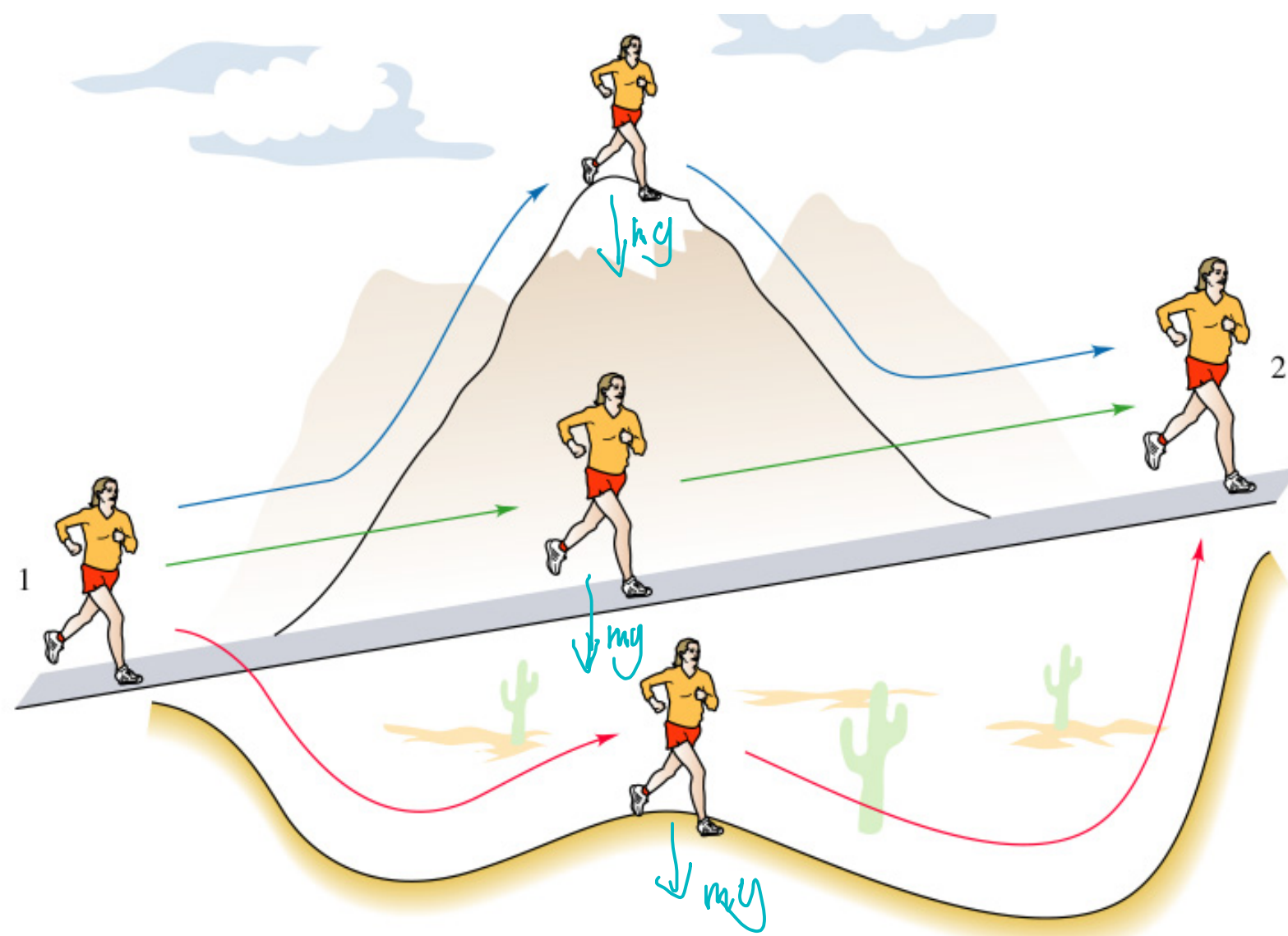
$W_F = -W_g = \Delta U_g = mgR(1 - \cos \theta)$



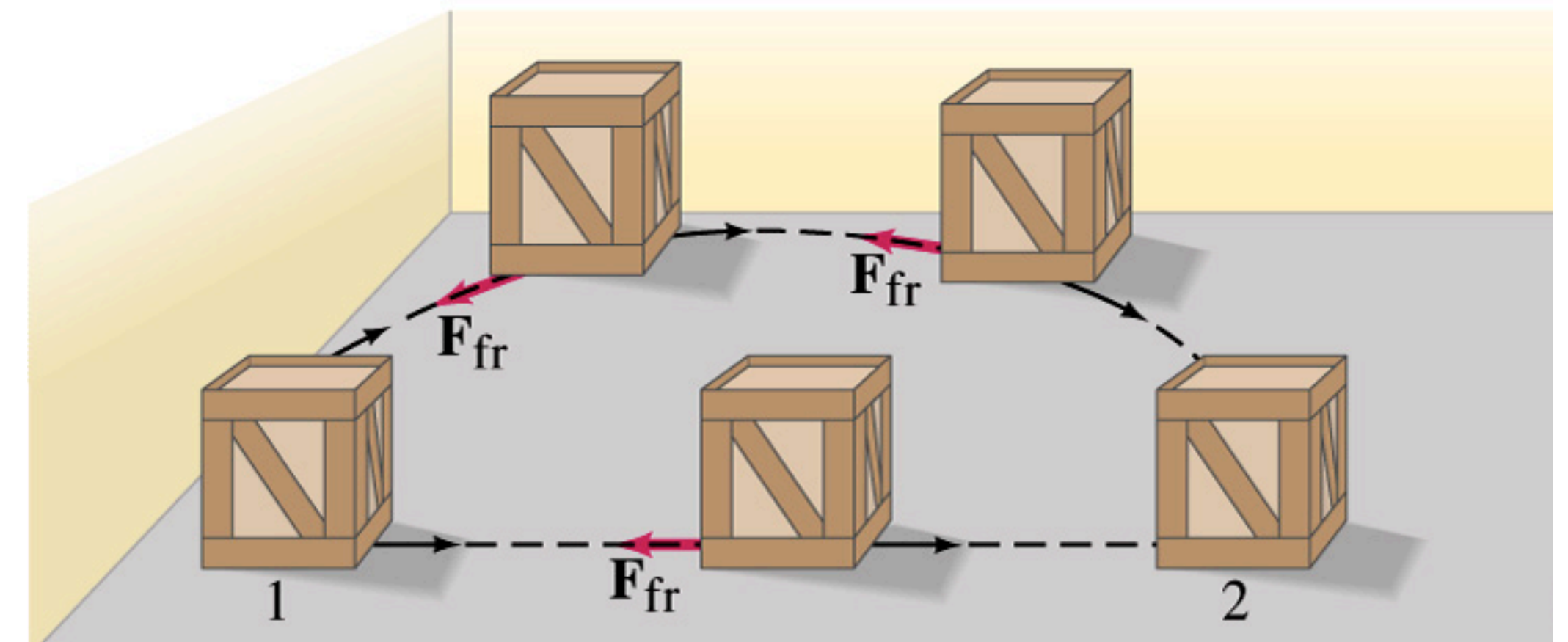
Conservative Force

- Def: work done by a conservative force **only** depends on the **initial** and **final** positions, and does not depend on the path.
 - Alternate. def.: work done by a conservative force moving around any closed path is zero.
- Conservative force: **gravity**, spring force, static electromagnetic force, ...
- Non-conservative force: friction, ...; it is also called dissipative force.
(reduce force)

$$W = -F \times \overset{\text{friction}}{\mu N} \cdot x < 0$$



Work done by gravity is the same no matter which path the athlete takes.



Work done by friction in a closed path is NOT zero.

Conservative Force

- A conservative force is usually associated with a potential energy U , such as gravity and gravitational potential energy.

- $F = -\frac{dU}{dx}$; In 3D it is $\vec{F} = -\vec{\nabla} U$, where $\vec{\nabla} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)$. not required

- Curl of conservative force $\vec{\nabla} \times \vec{F} = 0$.

potential

U_g U_{el}

Conservative force

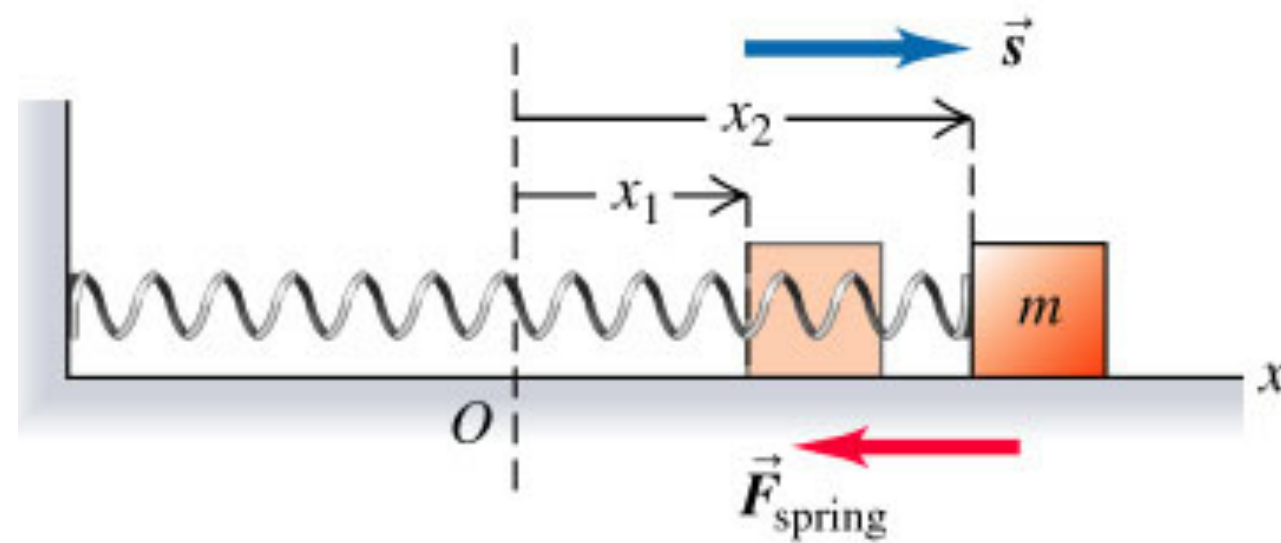
gravity spring force

$F = -\frac{dU}{dx} = -\frac{d(mgy)}{dy} = -mg$

$\left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right) \times \vec{F} = 0$

$\hookrightarrow \nabla \times (F_x\hat{i} + F_y\hat{j} + F_z\hat{k})$

Elastic Potential Energy



force and displacement opposite

- Hooke's law: $F = -kx$, where k the spring constant (unit: N/m)
- Elastic potential energy $U_{el} = \frac{1}{2}kx^2$
 - ▶ Elastic potential energy is **always non-negative**, but gravitational potential energy could be negative.

$$N = -[k] \cdot m$$

$$[k] \text{ N/m}$$

Diagram 1: Spring stretched from x_1 to x_2 . Force $F = -kx$.

Diagram 2: Spring stretched from x_1 to x_2 . Work done $W_{el} = \int_{x_1}^{x_2} -kx \cdot dx = -\frac{1}{2}kx^2 \Big|_{x_1}^{x_2}$

$$= -\left(\frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2\right)$$

$$= -\left(U_{el,2} - U_{el,1}\right)$$

$$= -\Delta U_{el}$$

Potential energy U $\xrightarrow{F = -\frac{dU}{dx}}$ Conservative force F

$W_F = -\Delta U_F$



Robert Hooke (1635 - 1703)

Conservation of Mechanical Energy

- Mechanical energy: kinetic energy, elastic potential energy, gravitational potential energy, ..
- If only conservative forces do the work during a process, then the mechanical energy is conserved!
 - $K + U_{el} + U_g = \text{const}$

$W \rightarrow K:$ $W_{all} = \Delta K$

$$W_{all} = W_g + W_{el} = -\Delta U_g - \Delta U_{el} = \Delta K$$

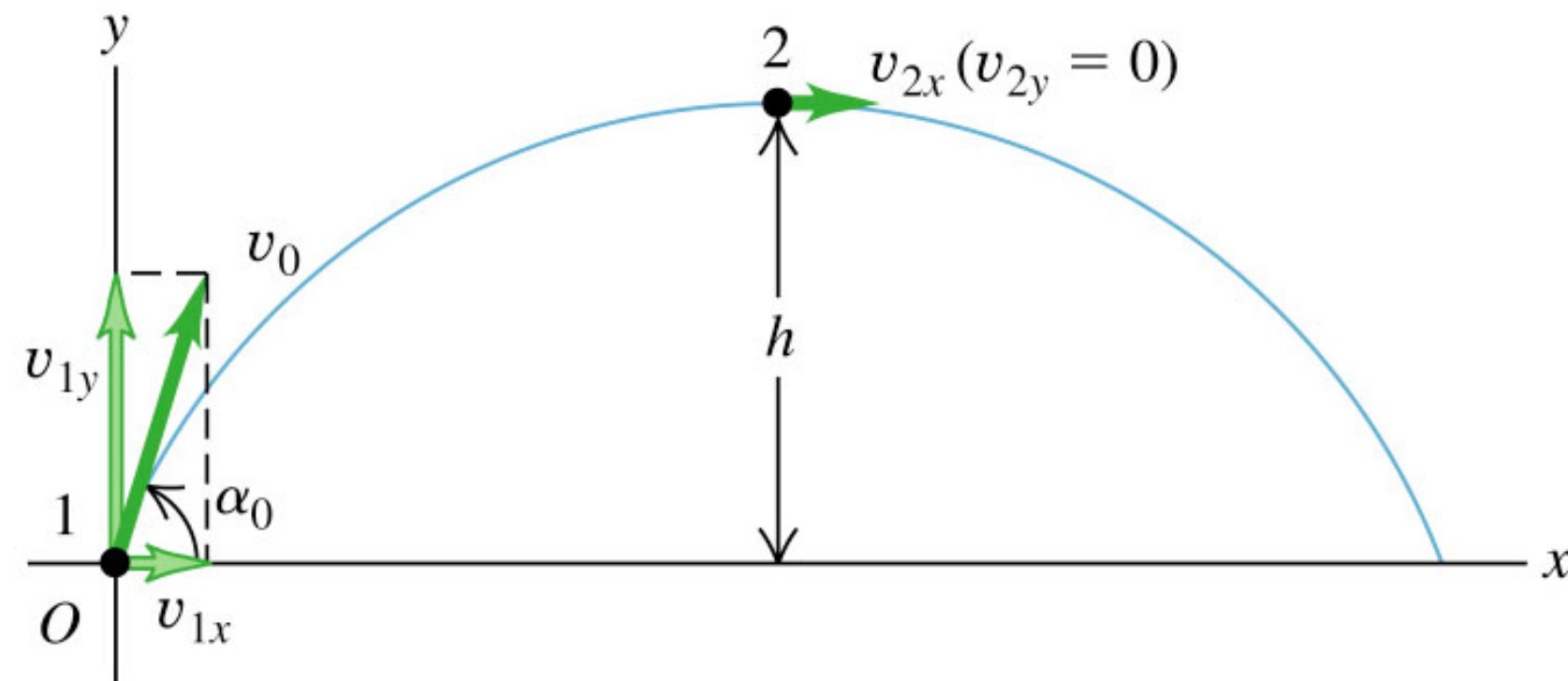
$$W_g = -\Delta U_g$$

$$\Delta K + \Delta U_g + \Delta U_{el} = 0$$

$$W_{el} = -\Delta U_{el}$$

$$K + U_g + U_{el} = \text{const.}$$

Revisit Projectile Motion



An object is projected at an initial speed v_0 with an angle α_0 w.r.t. horizontal. Use energy conservation to find the maximum height h of projectile motion.

$$(I) \quad v_{y,0} = v_0 \sin \alpha_0$$

$$2a(x-x_0) = v^2 - v_0^2$$

$$x = \frac{-v_0^2}{2a}$$

$$= \frac{v_0^2}{2g}$$

$$h = \frac{v_0^2 \sin^2 \alpha_0}{2g}$$

$$(II) \text{ initial } E_i = \frac{1}{2} m v^2 + 0$$

→

$$h=0 \quad \swarrow$$

$$\text{final } E_f = \frac{1}{2} m (v \cos \alpha)^2 + mgh$$

$$\frac{1}{2} m v^2 = \frac{1}{2} m (v \cos \alpha)^2 + mgh$$

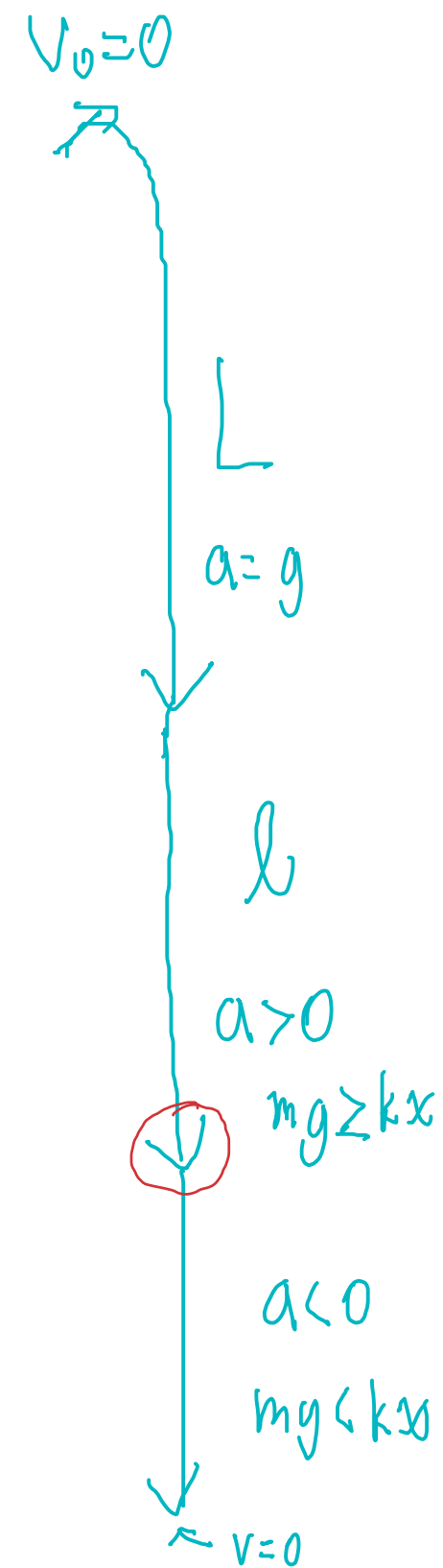
$$\frac{1}{2} m v^2 \sin^2 \alpha_0 = mgh$$

$$h = \frac{v^2 \sin^2 \alpha_0}{2g}$$

Bungee Jumping

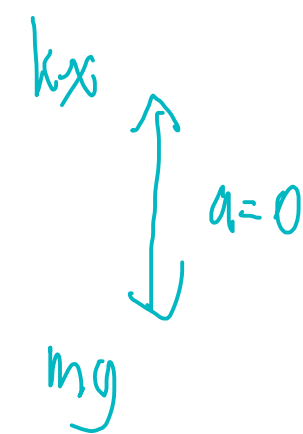
A person of mass m exploits bungee jumping at a cliff. The bungee cord has a length of L and a spring constant k . The person then jumps off the cliff with an initial velocity $v_0 = 0$.

a) Find the maximum velocity of the person during the bungee jumping.



a) $V_{max} \Leftrightarrow a = 0 \Leftrightarrow mg = kx$

ME: $0 + 0 + 0 = \frac{1}{2}mv^2 - mg(L + l) + \frac{1}{2}kl^2$



Dynamics: $mg - kl = 0$

$$l = \frac{mg}{k}$$

$$\frac{1}{2}mv^2 - mg\left(L + \frac{mg}{k}\right) + \frac{1}{2}\left(\frac{mg}{k}\right)^2 = 0$$

$$v = \sqrt{2gL + \left(1 + \frac{mg}{2kl}\right)}$$

Bungee Jumping

A person of mass m exploits bungee jumping at a cliff. The bungee cord has a length of L and a spring constant k . The person then jumps off the cliff with the cord tied, with an initial velocity $v_0 = 0$.

b) What is the maximum extension of the bungee cord?